Understanding IPv4 and IPv6 Addressing

The need for IP addressing

- An **Internet Protocol** (**IP**) address is a Layer 3 logical address that is assigned to all devices on a network to allow communication between nodes on different IP networks.
- The **Internet Assigned Numbers Authority** (**IANA**) is the governing body that is responsible for managing global **IP version 4** (**IPv4**) and **IP version 6** (**IPv6**) address assignments, **Autonomous System Number** (**ASN**) allocation to organizations that manage a large number of public networks, protocol assignment, and root **Domain Name System** (**DNS**) directories and services of the world.
- To assist with distributing public IP addresses to **Internet Service Providers** (**ISPs**) around the world, IANA delegated the responsibility of IP distribution to five **Regional Internet Registries** (**RIRs**) around the world.
- Each RIR is responsible for distributing IPv4 and IPv6 address blocks to specific regions and geolocation.

The following are the five RIRs and their responsible geolocations:

- **African Network Information Center** (**AFRINIC**): Supports the continent of Africa
- **Asia-Pacific Network Information Centre** (**APNIC**): Supports regions of Asia and the Pacific
- **American Registry for Internet Numbers** (**ARIN**): Supports regions of Canada, the USA and parts of the Caribbean
- **Latin America and Caribbean Network Information Centre** (**LACNIC**): Supports Latin America and parts of the Caribbean regions
- **Reseaux IP Europeens Network Coordination Centre** (**RIPE NCC**): Supports Europe, the Middle East and Central Asia

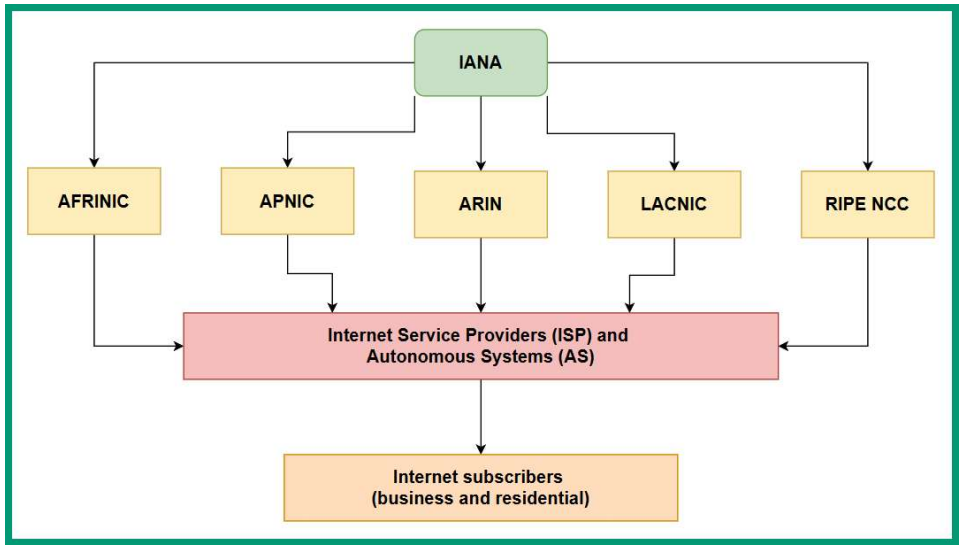


Fig. – IP network blocks delegation

The ISPs use their AS numbers to share their network routing prefixes with other ISPs using the **Border Gateway Protocol** (**BGP**).

Important note

BGP is an **Exterior Gateway Protocol** (**EGP**) that allows network operators such as ISPs to exchange their routing information with each other, allowing users on the internet to reach networks beyond their local ISP.

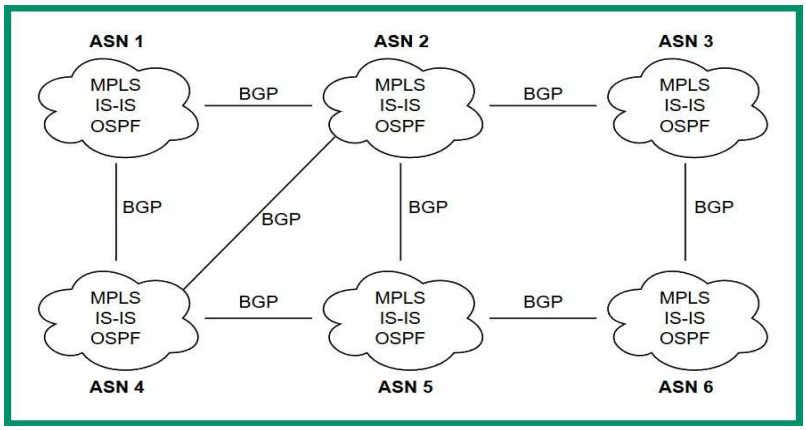


Fig. – Interconnected ISPs sharing network routes

- As shown in the preceding diagram, each network operator or ISP is assigned a unique AS number that allows them to interconnect and share their network routes with other ISPs while using BGP to share their routes.
- Internally, network operators use various **Interior Gateway Protocols** (**IGPs**) such as **Open Shortest Path First** (**OSPF**), **Intermediate System to Intermediate System** (**IS-IS**) and **Multiprotocol Label Switching** (**MPLS**) to route traffic within their networks.

  - As the internet began to grow quickly, IPv4 addresses for the internet were determined to one day be exhausted over time.
  - As early as 1999, the world began to see a new generation of logical addresses being deployed, known as IPv6 addresses.
  - These are 128-bit logical addresses that are written in hexadecimal notation.
  - IANA implemented the concept of creating two logical address spaces for IPv4 to help reduce the depletion of IPv4 addresses around the world.
    - Private Address Space
    - Public Address Space

Public versus private address spaces

- The **public address space** specified the IPv4 addresses that are routable on the internet and can be assigned to devices that are directly connected to the internet.
- The **private address space** specifies the classes and ranges of IPv4 addresses that are non-routable on the internet and can be assigned to devices on private networks only.
- To **help reduce the wastage of public IPv4 addresses, IANA created the private IPv4 address space**, which allows organizations to use a specific set of IPv4 addresses on private networks.
- If all devices within the private networks of organizations were configured with a unique public IPv4 address – the public address space would be exhausted a lot quicker than expected.
- The following table shows the public classes of IPv4 addresses:

Class	Range	Default Subnet Mask
A	1.0.0.0 – 126.255.255.255	255.0.0.0
B	128.0.0.0 – 191.255.255.255	255.255.0.0
C	192.0.0.0 – 223.255.255.255	255.255.255.0
D	224.0.0.0 – 239.255.255.255	N/A
E	240.0.0.0 – 255.255.255.255	

Fig. – IPv4 public address space

- As shown in the preceding table, classes A, B and C can be assigned to devices that are directly connected to the internet.
- The Class D address range is used for multicast communication between applications and services that operate on devices within a network
- The Class E address range is reserved for experimental uses.

Class	Range	Default Subnet Mask
A	10.0.0.0 – 10.255.255.255	255.0.0.0
B	172.16.0.0 – 172.31.255.255	255.255.0.0
C	192.168.0.0 – 192.168.255.255	255.255.255.0

Fig. – Private IPv4 address space

- As shown in the preceding table, the private IPv4 address spaces contain three classes of address ranges.
- Each class of address can be used within any organization's private network. This means two or more organizations can be using the same private IPv4 address classes within their private networks without causing any conflict.
- Before devices on a private IPv4 network can communicate with devices with public IPv4 addresses on the internet, the source private IPv4 address has to be translated into a public address.

Difference Between Private and Public IPv4 Addresses

Feature	Private IPv4 Address	Public IPv4 Address
Usage Scope	Used within local networks (LAN, home, office)	Used on the internet (globally routable)
Accessibility	Not reachable from the internet directly	Reachable over the internet
Uniqueness	Can be reused in multiple networks	Must be unique globally
Assigned By	Set manually or by DHCP within network	Assigned by ISP
Need for NAT	Yes — requires NAT to communicate with internet	No NAT needed for direct communication
Security	More secure by nature (isolated)	Less secure — exposed to the public internet
Cost	Free to use	Usually paid (provided by ISPs)
Example	192.168.1.10	8.8.8.8

Important note

The subnet mask is used to determine the network and host portion of an IP address, and the total number of IP addresses and usable IP addresses within a network.

Classful IP address

- Originally, the IPv4 address spaces used **classful** addresses, which describe how each IPv4 address class contains a specific number of networks.
- Here, each network contains a specific number of usable/assignable IPv4 Addresses to host devices.

The following table shows the IPv4 classful addressing information:

Class	Range	Default Subnet Mask	Number of Networks	Number of Usable IPV4 Addresses
A	1.0.0.0 – 126.255.255.255	255.0.0.0	126	16,777,214
B	128.0.0.0 – 191.255.255.255	255.255.0.0	16,384	65,534
C	192.0.0.0 – 223.255.255.255	255.255.255.0	2,097,152	254

Fig. – Classful IPv4 addresses

Disadvantages

- One of the **major issues with classful addressing** is the **wastage of unused IP addresses** within an organization.
- It was not flexible enough to support organizations of various sizes.
- The routers within service providers' networks contained very large routing tables, which affected the performance of a router.
- It was not flexible for businesses of different sizes.
- Any unused IPv4 addresses were not reallocated to another organization.

Classless IP address

- One solution was to implement **classless** addresses, **which removed the need to use a default subnet mask** for a specific range of IPv4 addresses.
- In this case, a Class A IPv4 address does not need to use the default subnet mask of 255.0.0.0 in classless addressing.
- It allows networking professionals to use custom subnet masks with any IPv4 address.

Difference Between Classful and Classless Addressing

Feature	Classful Addressing	Classless Addressing (CIDR)
Introduction	Early IP addressing method (used before 1993)	Introduced as CIDR (Classless Inter-Domain Routing) in 1993
IP Division Method	IPs are divided into fixed classes: A, B, C, D, E	No fixed classes, uses flexible subnet masks
Subnet Mask	Default masks only → Class A: /8, B: /16, C: /24	Custom masks allowed → e.g., /10, /27, /30
Address Utilization	Wastage of IPs due to rigid block sizes	Efficient IP usage by variable-length subnetting
Routing	Uses *classful routing protocols* like RIPv1, IGRP	Uses *classless routing protocols* like RIPv2, OSPF, EIGRP, BGP
Example Network	192.168.0.0 (Always /24)	192.168.0.0/26 (Can be customized)
Support for VLSM (Variable Length Subnet Masking)	Not allowed	Allowed
Status Today	Obsolete (not used on the internet anymore)	Fully adopted & recommended

Network Address Translation (NAT)

- It allows a private IPv4 source address to be translated into a public IPv4 address and vice-versa.
- It allows devices on a private network to communicate with devices on a public network.
- To help ensure devices on a private IPv4 network can communicate with the internet, a NAT-enabled router is implemented between the networks.

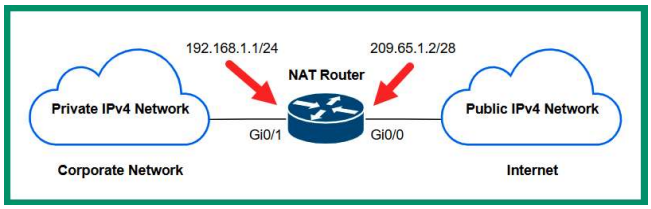


Fig. – NAT router

- There's a private IPv4 network on the left that indicates an organization's private network while on the right there's the internet, where all devices are assigned a public IPv4 address.
- To interconnect these two different networks, a router is implemented to route traffic between the different networks.
- To ensure devices on the private network can communicate with devices on the internet, NAT is configured on the router to translate any source IPv4 address that originates from the private network to the public IPv4 address that's configured on the router.
- All outbound traffic from the corporate network to the internet will be assigned a new source IPv4 public address of 209.65.1.2.
- This allows devices on the internet to see the source traffic as originating from a device with a public IPv4 address of 209.65.1.2 and not from the private IPv4 network.

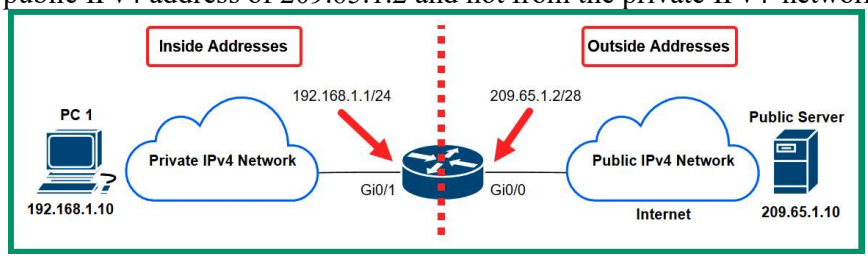


Fig. – NAT operations

- The NAT-enabled router/modem identifies the addresses as **inside** and **outside**.
- The inside addresses are the private IPv4 addresses that are to be translated by the NAT-enabled router/modem. The outside addresses are those that are seen by all devices on the public address space – the internet.
- *PC1* is connected to a private network within an organization having a private IPv4 address of 192.168.1.10. This address is referred to as the Inside Local address. If *PC 1* attempts to send a message to the *Public Server* on the internet, the packets from *PC 1* will be restricted from entering the server provider's network and the internet.

The following steps explain how NAT works:

1. *PC 1* creates a packet with the source IPv4 address of 192.168.1.10 (*Inside Local*) and a destination address of 209.65.10 (*Outside Global*), and forwards the message to the default gateway (router) at 192.168.1.1.
2. The router inspects the source and destination IP addresses within the IP header of the packet to determine the destination and checks its routing table for a route (path).
3. Before the router forwards the packet to the internet, it translates the source private IPv4 address from 192.168.1.10 to the public IPv4 address on the router as 209.65.1.2 (*Inside Global*), then sends the packet to the destination server.
4. The *Public Server* will see the source IPv4 address of the packet as 209.65.1.2 (router) but not *PC 1* because NAT allows the private network to be hidden behind the public IPv4 address of the router.

Advantages of using NAT on a network:

- It helps conserve the public IPv4 address space.
- It allows an entire organization's private network to be hidden behind a single public IPv4 address.
- Since private IPv4 addresses are non-routable on the internet, organizations can use any private IPv4 address blocks on their private, internal networks.

Disadvantages of NAT:

- Network performance is affected. As traffic is sent to the router or modem to be translated before it is placed on the network, there is some delay as the router or modem has to perform the actual translation process.
- Since NAT modifies the IPv4 addresses on the packet, **Virtual Private Network** (**VPN**) solutions that use **IP security** (**IPsec**) to establish a secure logical tunnel over an unsecure network does not work well.
- End-to-end connectivity is lost between a sender and receiver. This makes it difficult for a receiver to determine the true source of a message.

The following are the three common types of NAT:

- Static NAT
- Dynamic NAT
- Port address translation (PAT)

Static NAT

Static NAT provides a permanent one-to-one mapping between a private (inside local) and a public (inside global) IP address.

Characteristics

- Mapping is fixed and manually configured.
- Same public IP is always assigned to the same internal host.
- It is useful for devices that must be accessible from the Internet (e.g., servers, CCTV, web or mail servers).

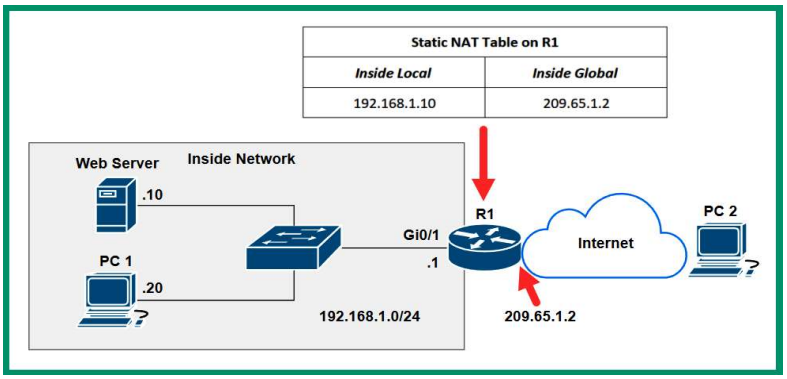


Fig. – Static NAT

☑ **Advantages**

- Simple and predictable.
- Allows hosting internal servers with fixed public addresses.

✗ **Disadvantages**

- Requires **one public IP per internal host**.
- Not scalable for large networks.

Dynamic NAT

Dynamic NAT provides a **temporary one-to-one mapping** between private and public IPs from a **pool** of public addresses.

Characteristics

- Mappings are assigned **automatically** from a pool when needed.
- It is released when the session ends.
- It works only if a public IP is available in the pool.

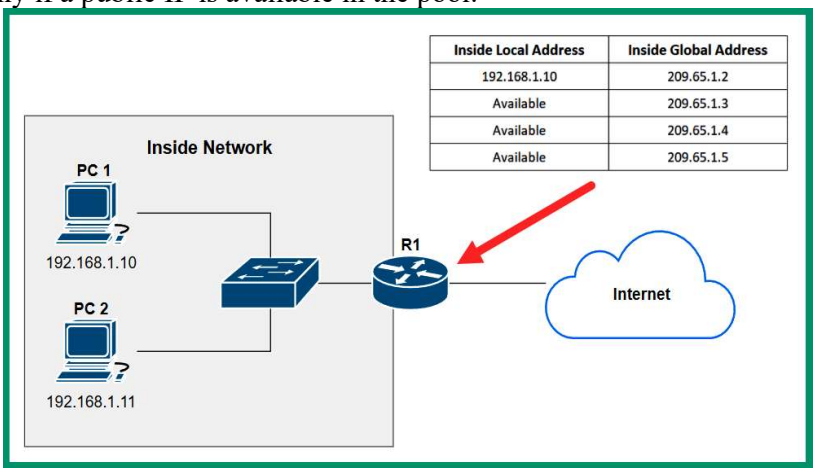


Fig. – Dynamic NAT

☑ Advantages

- It is more flexible than Static NAT.
- Efficient use of a limited public IP pool.

✗ Disadvantages

- It requires multiple public IPs.
- It is not ideal for large networks or home users.

Port address translation (PAT)

PAT allows **multiple private IP addresses** to share **a single public IP address** by using **different port numbers** to identify sessions.

Characteristics

- Many-to-one mapping.
- Router uses unique **TCP/UDP port numbers** to differentiate sessions.
- Most common form of NAT (used in home routers).

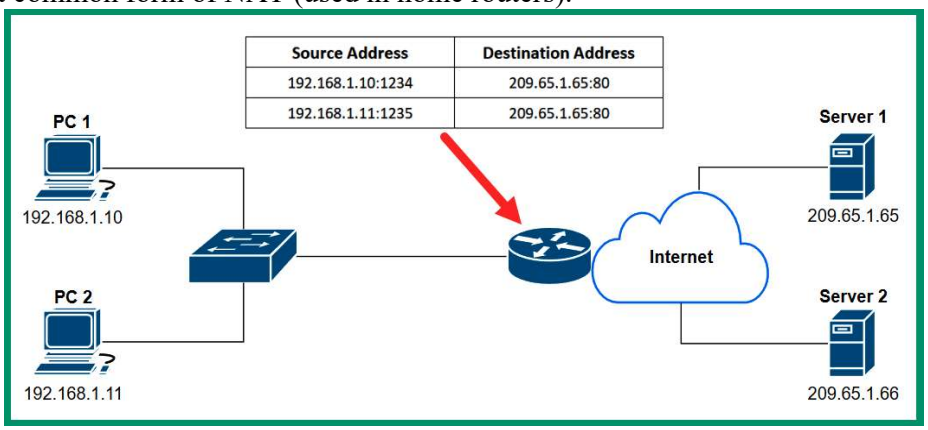


Fig. – Port address translation

☑ **Advantages**

- Very high scalability.
- Only one public IP is needed for the entire network.
- It provides basic security (internal IPs hidden).

❌ **Disadvantages**

- Uses port numbers for differentiation, so some protocols (like IPsec) need special handling.

Exploring the structure of IPv4 and IPv6

Here you will learn how to convert IPv4 addresses from binary into decimal and vice versa, and explore various address types and their use cases on a network.

Fundamentals of IPv4

- Within an IPv4 address, four octets are separated by a dot (.), and each octet is made up of 8 bits.
- Therefore, 4 octets x 8 bits per octet = 32 bits in length per IPv4 address. The following is an example of an IPv4 address in dotted-decimal and binary notation:

	1st Octet	2nd Octet	3rd Octet	4th Octet
Dotted Decimal	192	168	10	20
Binary	1100 0000	1010 1000	0000 1010	0001 0100

Fig. – IPv4 address format

Converting binary into decimal

Let's take a further look into the format of an IPv4 address with its binary format. Since an IPv4 address is 32 bits in length with four octets, the following is an example of an IPv4 address in binary notation:

11000000.10101000.00000001.10000001

Now let's apply the theory to converting the IPv4 address of 11000000.10101000.00000001 .10000001 into a dotted-decimal notation. To perform this exercise, follow these steps:

1. Let's start with the first octet and place its values within the table, as shown here:

Radix	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal	128	64	32	16	8	4	2	1
Binary	1	1	0	0	0	0	0	0

Fig. – Converting the first octet

2. Using the same principle from the previous step, let's assign the values of the second octet within the positioning system to determine its decimal value:

Radix	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal	128	64	32	16	8	4	2	1
Binary	1	0	1	0	1	0	0	0

Fig. – Converting the second octet

3. Next, let's convert the third octet by placing its bit values within the positioning systems, as shown in the following table:

Radix	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal	128	64	32	16	8	4	2	1
Binary	0	0	0	0	0	0	0	1

Fig. – Converting the third octet

4. Next, let's convert the fourth octet by placing its bit values within the positioning system once more:

Radix	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal	128	64	32	16	8	4	2	1
Binary	1	0	0	0	0	0	0	1

Fig. – Converting the fourth octet

5. The final step is simply putting everything all together: 11000000.10101000.0000000 1.10000001 = 192.168.1.129.

Converting decimal into binary

- Here, you will learn how to use an eight-step method to convert an IPv4 address, 172.19.43.67, from decimal into binary, one octet at a time.
- Let's start by converting the first octet, 172, from decimal into binary.

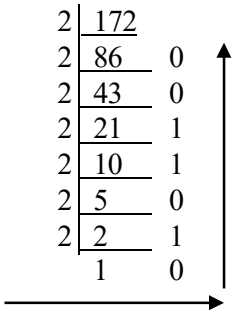


Fig. – Converting 172 into binary

- We get the result as 10101100.
- Similarly, converting the 2nd octet 19 into binary we will get the value 00010011, 3rd octet 43 into binary we will get 00101011 and 4th octet 67 into binary we will get 01000011.

Lastly, let's put everything all together and view the binary notation of 172.19.43.67:

	1st Octet	2nd Octet	3rd Octet	4th Octet
Decimal	172	19	43	67
Binary	10101100	00010011	00101011	01000011

As shown in the preceding table, the IPv4 address is 172.19.43.67 = 10101100.0001001 1.00101011.01000011.

Fundamentals of IPv6

An IPv6 address consists of 128 bits in length and is written in hexadecimal notation. Hexadecimal contains both numbers and letters in the following range:

0 1 2 3 4 5 6 7 8 9 A B C D E F

This 128-bit address allows IPv6 to scale to 2^{128} IPv6 addresses, which allows 3.4 x 10^{38} IPv6 addresses that can be uniquely assigned to devices.
Each IPv6 address is made up of 128 bits that are grouped into eight (8) hextets. Each hextet contains 16 bits, so 8 hextets x 16 bits per hextet = 128 bits in length per IPv6 address.

The following is an example of an IPv6 address:

2001:0DB8:0000:1111:0000:0000:0000:0200

Let's break down the following IPv6 address into a more simplified version: First, all leading

0s in a hextet can be removed.
When there are two or more hextets that are all zeros, a double colon (::) is used to replace the consecutive zeros within the IPv6 address.
The following is a simplified version of the original IPv6 address:

2001:DB8:0:1111:0:0:0:200

Then, replace consecutive zero-only hextets with a double colon (::). The following is a further simplified version:

2001:DB8:0:1111::200

The default subnet mask or prefix length of an IPv6 address is /64. This means the first 64 bits of the IPv6 address represent the network portion of the address and the second portion represents the interface ID.

The following diagram shows the parts of an IPv6 address:

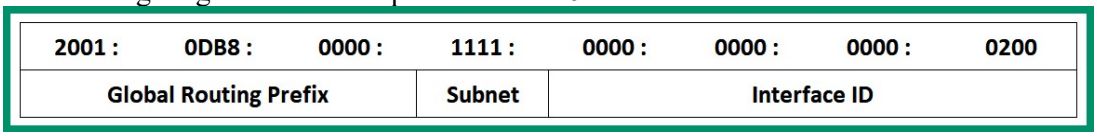


Figure 4.22 – IPv6 address structure

- The first three hextets represent the **Global Routing Prefix** portion, which contains the first 48 bits of the address. This portion of the IPv6 address is assigned by the service provider such as the ISP.
- The fourth hextet (16 bits) represents the **Subnet** ID, which is used by the ISP to create subnetworks of the network block.
- The last 64 bits represent the **Interface ID** portion. Combining the Global Routing Prefix, Subnet ID, and Interface ID, a client is assigned a unique 128-bit IPv6 address on its network interface card.

Types of IPv4 and IPv6 addresses

There are various types of IPv4 and IPv6 addresses and it's important to understand them as an aspiring network professional.

Automatic Private IP Addressing (APIPA)

There are many reasons a client device may not receive an IP address from the network, as follows:

- The client is unable to communicate with the DHCP server on the network
- The client is configured to use a static IP address that does not change
- The DHCP server is not present or offline on the network
  - By default, many client devices are configured to automatically communicate with a DHCP server and receive an IP address from it.
  - However, if a client is unable to reach the DHCP server on the network, the client will automatically assign itself a special unique IPv4 address ranging from 169.254.0.1 to 169.254.255.254 with a default subnet mask of 255.255.0.0. This is a feature known as Automatic Private IP Addressing (APIPA), which is built into many operating systems such as Microsoft Windows.

Important note

On an IPv4 network, the APIPA address is sometimes referred to as an IPv4 **Link-Local** address. However, APIPA is a Microsoft-specific term for "Link-Local," which is preferred for Linux-based operating systems.

Extended unique identifier (EUI-64)

- On some IPv6 networks, a stateless IPv6 technology known as **Stateless Address Autoconfiguration** (**SLAAC**) is implemented that helps clients obtain global unicast IPv6 addresses on the network.
- One of the main differences between SLAAC and DHCPv6 is that SLAAC does not keep a record of each client's IPv6 address assignment or details.
- It provides the network portion of the IPv6 address to the client – that is, the first 64 bits of the IPv6 address.
- The client uses a process known as **Extended Unique Identifier 64** (**EUI-64**) to convert its 48-bit **Media Access Control** (**MAC**) address into a 64-bit address, which will become the Interface ID portion of the IPv6 address.
- This allows the client to combine the 64-bit network prefix and the newly created EUI-64 address to create the 128-bit IPv6 address for the network adapter.
- Let's imagine a computer is connected to an IPv6 network and is seeking a DHCPv6 server.

1. However, SLAAC is enabled on the network and provides the client with 2001:DB8:0:1111::/64, the network portion of the IPv6 address for its network adapter.

- The 48-bit MAC address is used on the computer's network adapter and the EUI-64 process is used to create the 64-bit interface ID portion of a 128-bit IPv6 address.

The following steps describe the EUI-64 process of converting a 48-bit MAC address of a network adapter to create a 64-bit address that will be used as the interface ID portion to create a unique IPv6 address:

1. First, split the 48-bit MAC address into half, separating the **Organizational Unique Identifier** (**OUI**) and the device portions, as shown in the following diagram:

FC	99	47	75	CE	E0
11111100	10011001	01000111	01110101	11001110	11100000

2. Next, insert the hexadecimal value, FFFE, in the middle of the 48-bit MAC address, as shown in the following diagram:

FC	99	47	FF	FE	75	CE	E0
11111100	10011001	01000111	**11111111**	**11111110**	01110101	11001110	11100000

3. Next, flip the seventh bit within the first byte so that a 0 will become a 1 or a 1 will become a 0. This bit indicates if the NIC is administered locally (0) or is globally unique (1), as shown in the following diagram:

11111110	10011001	01000111	**11111111**	**11111110**	01110101	11001110	11100000

4. Next, convert the binary into hexadecimal to view the EUI-64 portion of the address, as shown in the following diagram:

FE	99	47	FF	FE	75	CE	E0

5. Lastly, putting it all together, the EUI-64 bit IPv6 address that will be assigned to the device's
network adapter is 2001:DB8:0:1111:FE99:47FF:FE75:CEE0.

Unicast

- It can be IPv4 and IPv6 address which is uniquely assigned to the network adapter of a device.
- It allows **one-to-one** communication between a sender and receiver device over a network.
- The following diagram shows an example of a unicast network transmission between two devices:

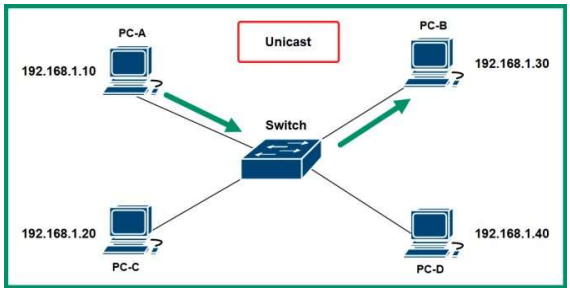


Fig. – Unicast communication

Multicast

- It exists within both the IPv4 and IPv6 address spaces.
- It allows **one-to-many** communication between devices on a network.
- Enterprise-grade routers within a large organization are usually configured with a dynamic routing protocol, which allows each router to automatically learn new networks and maintain an up-to-date routing table by exchanging routing information between themselves.
- These routers send and receive messages to a multicast address group, which is only used by devices running the same dynamic routing protocol.

The following diagram shows an example of multicast transmission over a network:

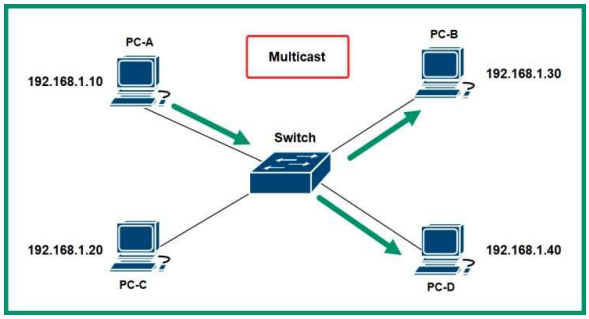


Fig. – Multicast communication

Broadcast

- It allows **one-to-all** communication over an IPv4 network.
- It is only applicable to IPV4 addresses.
- A computer that's connected to a network can send a single message to the network's broadcast address, which allows all devices within the same IP network to receive the message from the sender.
- Within a network that uses a Network ID of 192.168.1.0 and has a subnet mask of 255.255.255.0, the broadcast IP address will be 192.168.1.255.

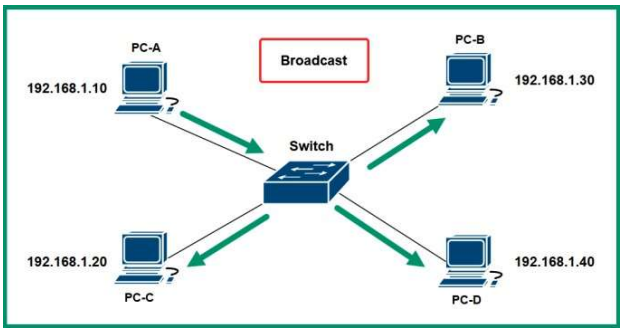


Fig. – Broadcast communication

Anycast

- These are used for **one-to-nearest** communication between devices over a network.
- It is only applicable only to IPV6 addresses.
- It allows the same unicast IPv6 address to be shared between multiple devices, such as servers on the internet.
- When a client such as a computer sends a message to an anycast IPv6 address, the message is delivered to the geographically nearest server that's configured with the anycast address.

Delving into IPv6 concepts

- To help ensure devices that exist on both IPv4 and IPv6 networks can communicate with each other, various IPv6 technologies allow both versions of IP to coexist.

Tunneling

- It allows IPv6 packets to be transported over an IPv4 network.

- The forwarding router is responsible for encapsulating the IPv6 packet inside an IPv4 packet before sending it over the IPv4 network.
- This type of tunneling is referred to as 6to4 tunneling.
- 4to6 tunneling allows an IPv4 packet to be encapsulated within an IPv6 packet so that it can be transported over an IPv6 network.

Dual stack

- Dual stacking both IPv4 and IPv6 addresses on the same NIC of an end device or interface of a networking device such as a router will allow the device to communicate efficiently on both IPv4 and IPv6 networks.
- If the device has sent a message to a host on an IPv4 network, it will use the IPv4 address that's configured on its local NIC or interface.
- If the device has to send a message to a host on an IPv6 network, it will use the IPv6 address that's configured on the local NIC or interface.

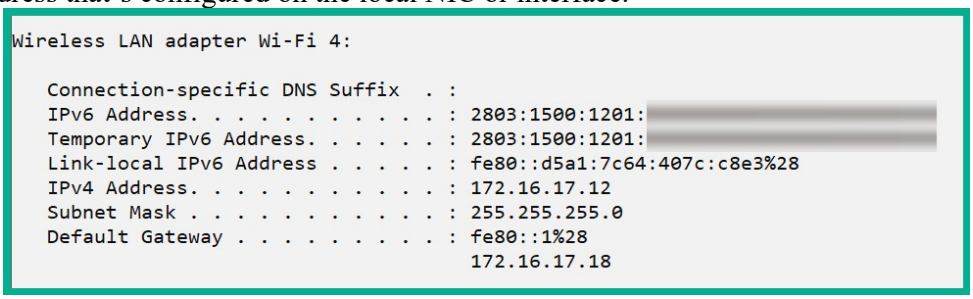


Fig. – Dual stack NIC

Translation

- **Network Address Translation 64** (**NAT64**) allows devices within an IPv6 network to communicate with hosts on an IPv4 network using an address translation.
- NAT64 is configured on routers to translate IPv6 addresses into IPv4 addresses, which allows IPv6-enabled devices to communicate with IPv4 hosts and vice versa.

Router advertisement

- Devices on an IPv6 network automatically obtain a GUA IPv6 address using ICMPv6 messages.
- Whenever an IPv6-enabled device such as a computer is connected to an IPv6 network, it sends a Router Solicitation (RS) message to the network to discover any IPv6 routers.
- An IPv6-enabled router responds with a Router Advertisement (RA) message, which is used to inform the host on the network how to obtain a GUA IPv6 address.

Furthermore, the RA message provides the following network information to the client:

- The network prefix and the prefix length of the address
- The default gateway IPv6 address for the network
- The DNS server IP addresses and domain name

Additionally, the RA messages provide the following methods for configuring a GUA IPv6 address for a client on the network:

- Stateless Address Autoconfiguration (SLAAC)
- SLAAC with stateless DHCPv6
- Stateful DHCPv6

Stateless Address Autoconfiguration (SLAAC)

- It allows devices on a network to be configured with a GUA IPv6 address without the need for a Dynamic Host Configuration Protocol v6 (DHCPv6) server.

- The following is the process of a client obtaining a GUA IPv6 address using SLAAC on an IPv6 network:
  1. A client on the network sends an RS message to seek any IPv6-enabled routers.
  2. The IPv6-enabled router responds with an ICMPv6 RA message and provides the network prefix and the prefix length.
  3. The client uses the EUI-64 process to convert its 48-bit MAC address on the local NIC to create a 64-bit Interface ID. The 64-bit Interface ID is appended to the end of the 64-bit network prefix to create a 128-bit GUA IPv6 address for the client.

SLAAC with stateless DHCPv6

- Additionally, the RA message from the router can indicate to the client to use both SLAAC and stateless DHCPv6 to obtain a GUA IPv6 address. In this situation, the RA messages inform the client device of the following instructions:
  1. First, use SLAAC to create its own GUA IPv6 address.
  2. Second, use the router's IPv6 Link-Local address as the default gateway for the network. The router's IPv6 Link-Local addresses are set as the source addresses within the RA message from the router to the client.
  3. Lastly, use the stateless DHCPv6 server to obtain the DNS server addresses and domain names only. The stateless DHCPv6 server does not provide the IPv6 address, prefix length, or the default gateway.

Stateful DHCPv6

A stateful DHCPv6 server has similar functionalities to a traditional DHCP server on an IPv4 network as it provides the following configurations to clients:

- GUA IPv6 address
- Prefix length
- DNS server addresses
- Domain name

While using a stateful DHCPv6 server on a network, the RA messages from the router provide the default gateway address to clients. The router's IPv6 Link-Local address is included within the RA message as the source address.

Difference Between IPv4 and IPv6

Feature	IPv4	IPv6
Full Form	Internet Protocol version 4	Internet Protocol version 6
Address Size	32-bit	128-bit
Address Format	Dotted decimal notation (e.g., 192.168.1.1)	Hexadecimal, colon-separated (e.g., 2001:0db8:85a3::8a2e:0370:7334)
Number of Addresses	~4.3 billion	≈ 3.4×10^{38} addresses (virtually unlimited)
Configuration	Manual or DHCP	Auto-configuration (SLAAC) + DHCPv6
Security (IPSec)	Optional	Mandatory
Broadcast Support	Yes (broadcast address)	No broadcast — replaced by multicast and anycast
Fragmentation	Done by sender and routers	Done only by sender (routers don't fragment)
Checksum	Present	Removed for performance
QoS / Traffic Class	Limited (Type of Service field)	Improved (Flow Label field)
NAT (Network Address Translation)	Commonly used due to limited addresses	Not needed — vast address space